

Abhishek Nannuri

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Professional Summary

Robotic Software Engineer bridging classical autonomous systems with modern Physical AI, Strong background in C++, Python, and ROS2, with a focus on building reliable robotic manipulation systems in real-world settings. My work centers on Reinforcement Learning – specifically online off-policy and offline methods, applied to improve reliability in physical systems. I explore Residual RL to push Behavior Cloning policies from partial success towards consistent task completion. Worked with models such as Vision-Language-Action systems, Action Chunking Transformers, and diffusion policies, combining advances in AI with practical engineering to build systems that actually work in the real world.

Experience

Working Student & Master's Thesis	BMW	<i>München, De</i>	09/2025 - Present
• Research and develop Physical AI solutions for industrial robotics, leveraging Vision-Language-Action (VLA) models, Action Chunking Transformers (ACT), and Diffusion Policies (DP). • Design and implement Residual RL frameworks to fine-tune base Behavior Cloning (BC) policies, boosting success rates from 80% to 100% for real-world manipulation tasks. • Train and deploy online and offline off-policy Actor-Critic RL variants (SAC, TD3, RLPD, DSRL) and offline RL methods (DiffQL) for sample-efficient robotic control. • Developing vision-based reward functions to make RL and Residual RL more sample efficient for real world robotics.			
Intern Next Generation Robotics	BMW	<i>München, De</i>	03/2025 - 08/2025
• Develop and implement advanced algorithms for industrial robotics applications using modern C++, Python, and ROS2. • Engineer and optimize robotic manipulation solutions using MoveIt Pro and Behavior Trees for Universal Robot and Kuka platforms. • Conduct real-world testing and validation of algorithms on physical robotic manipulators in lab environments. • Contribute to DevOps workflows by developing CI pipelines and Docker configurations to streamline development processes.			
DevOps Engineer	Accenture	<i>Hyderabad, India</i>	05/2021 - 09/2022
• Collaborated on infrastructure modernization by assisting with AWS migration and implementing security protocols as part of the DevOps team. • Built CI/CD pipeline integration across Azure, AWS, and SharePoint, improving deployment efficiency by 40%. • Assisted in managing secure credential handling through Azure Key Vault and contributed to infrastructure optimization initiatives. • Participated in an enterprise application development team, working with C#, and PostgreSQL to enhance client acquisition capabilities.			

Publications

- **Adaptive Global Navigation for Sidewalk Robots: Integrating OSM Lanelets with Real-Time Pavement Segmentation-Based Correction**
 - Adapted the Lanelet2 HD map framework for autonomous last-mile delivery robots, translating OpenStreetMap (OSM) geographic coordinates into local UTM systems for precise global path planning on sidewalks.
 - Engineered a real-time, vision-based waypoint correction module using depth cameras and semantic segmentation to dynamically adjust global paths, preventing hazardous pavement edge drifting.
- **Multilayer Environment Model for Outdoor Autonomous Mobile Robot Navigation**
 - Contributed to the development of an extended three-layer High-Definition (HD) map architecture designed specifically for outdoor autonomous mobile robots.
 - Established a scalable, map-aware infrastructure that facilitates reliable global navigation and localization in complex urban environments.

Education

Master of Engineering [Hochschule Schmalkalden](#) Schmalkalden, DE 10/2022 - Present

- Major in Mechatronics & Robotics
 - **Thesis:** Vision-Based Dense Rewards for Finetuning Manipulation Policies via Reinforcement Learning
 - **Focus:** Designing dense reward functions to accelerate RL convergence and sample efficiency for industrial manipulation tasks

Bachelor of Technology [Karunya University](#) Coimbatore, IN 07/2017 - 05/2021

- Major in Mechanical Engineering

Projects

- **AMR Logistic System: Intracity Delivery Autonomous Mobile Robot for Urban Navigation System:** 08/2024 - Present
 - Engineered a production-grade Global Path Planner in C++ using Lanelet2 and GPS-based navigation featuring fault-tolerant operation and seamless ROS2 integration.
 - Architected comprehensive modular ROS2-based system architecture with robust exception handling and logging, significantly streamlining development workflow and system maintenance.
 - Implemented Local Trajectory Planner using Quintic Bézier splines achieving C2/G2 path continuity, reducing trajectory deviations by 40%.
 - Integrated state-of-the-art velocity profiling system using Dynamic Window Approach (DWA), Enhanced to satisfy robot's kinematic and dynamic constraints.
 - Crafted a custom RPROP optimization framework with gradient-based mapping, reducing trajectory traversal time by 25% compared to human-defined centerlines.
- **Visual Odometry for Localization in Autonomous Driving:** 04/2024
 - Programmed a feature extraction pipeline using SIFT descriptors and FLANN-based matching, reducing outliers by 85% in sequential image processing.
 - Implemented visual odometry using Essential Matrix Decomposition, achieving 92% accuracy in camera motion estimation.
 - Reconstructed vehicle trajectory by computing incremental camera pose transformations.
- **State Estimation and Localization for Self-Driving Car:** 03/2024
 - Formulated Extended Kalman Filter (EKF) sensor fusion pipeline integrating pre-recorded and processed IMU, GNSS, and LIDAR data for vehicle state estimation and localization.
 - Developed robust measurement update functions and error state modeling to process multi-sensor data with covariance tracking.
 - Built 3D visualization tools using Matplotlib for trajectory analysis and error validation across position and orientation states.
- **Road Lane Detection System - Computer Vision Based Advanced Driver Assistance:** 09/2023 - 12/2023
 - **Classical CV Pipeline:** Built an OpenCV/NumPy pipeline using Canny edge detection, Hough Transforms, and HSV/dynamic ROI masking; achieved 95% lane detection accuracy and reduced false positives by 40%.
 - **Sliding Window & Perspective Geometry:** Implemented Bird's-Eye-View perspective matrices, HLS color space, and polynomial curve fitting with centroid tracking, boosting curved road detection accuracy by 55%.
 - **Deep Learning (FCN-8):** Architected a Fully Convolutional Network (FCN-8) with a VGG16 backbone in TensorFlow/Keras for semantic road segmentation; optimized training using the tf.data API with batch prefetching.
 - **Performance:** Achieved 93% segmentation accuracy on the KITTI dataset and engineered a fail-safe execution loop that delivered real-time video processing at 60 FPS.

Technical Skills

- **Robotics & Control:** Reinforcement Learning (SAC, TD3, RLPD, DSRL, DiffQL), Residual RL, Vision-Language-Action (VLA), Action Chunking Transformers (ACT), Diffusion Policies, Behavior Cloning, Motion Planning, Inverse Kinematics
- **Tools & Simulation:** PyTorch, ROS2, MoveIt Pro, Behavior Trees, MuJoCo, Gazebo, OpenCV, Jax, Weights&Biases
- **Infrastructure:** Linux, Git, Docker, CI/CD, Cloud (AWS, Azure)